

A perspective on the future of transportable optical lattice clocks

Cite as: Appl. Phys. Lett. **120**, 140502 (2022); <https://doi.org/10.1063/5.0087894>

Submitted: 10 February 2022 • Accepted: 24 March 2022 • Published Online: 07 April 2022

 M. Takamoto,  Y. Tanaka and  H. Katori



View Online



Export Citation



CrossMark

ARTICLES YOU MAY BE INTERESTED IN

Frequency-stabilized Faraday laser with 10^{-14} short-term instability for atomic clocks
Applied Physics Letters **120**, 141102 (2022); <https://doi.org/10.1063/5.0083390>

Progress toward picosecond on-chip magnetic memory
Applied Physics Letters **120**, 140501 (2022); <https://doi.org/10.1063/5.0083897>

A proof-of-concept model of compact and high-performance ^{87}Sr optical lattice clock for space
AIP Advances **11**, 125116 (2021); <https://doi.org/10.1063/5.0064087>

 QBLOX



1 qubit

Shorten Setup Time
Auto-Calibration
More Qubits

Fully-integrated
Quantum Control Stacks
Ultrastable DC to 18.5 GHz
Synchronized <<1 ns
Ultralow noise



100s qubits

[visit our website >](#)

A perspective on the future of transportable optical lattice clocks

Cite as: Appl. Phys. Lett. **120**, 140502 (2022); doi: [10.1063/5.0087894](https://doi.org/10.1063/5.0087894)

Submitted: 10 February 2022 · Accepted: 24 March 2022 ·

Published Online: 7 April 2022



View Online



Export Citation



CrossMark

M. Takamoto,^{1,2} Y. Tanaka,³ and H. Katori^{1,2,4,a)}

AFFILIATIONS

¹Quantum Metrology Laboratory, RIKEN, Wako, Saitama 351-0198, Japan

²Space-Time Engineering Research Team, RIKEN, Wako, Saitama 351-0198, Japan

³Department of Earth and Planetary Science, Graduate School of Science, The University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan

⁴Department of Applied Physics, Graduate School of Engineering, The University of Tokyo, Bunkyo-ku, Tokyo 113-8656, Japan

^{a)} Author to whom correspondence should be addressed: hkatori@riken.jp

ABSTRACT

The unprecedented stability and accuracy of optical atomic clocks extend their role not only in frequency metrology but also in fundamental physics and geodesy. In particular, excellent stability of optical lattice clocks accessing a fractional uncertainty of 10^{-18} in less than an hour opens a new avenue for chronometric leveling, which resolves a height difference of one cm in a short averaging time. However, for field use of such clocks, there remains a challenge in developing a transportable system that can operate outside the laboratory. In this Perspective, we describe transportable optical lattice clocks and discuss their future applications to chronometric leveling.

Published under an exclusive license by AIP Publishing. <https://doi.org/10.1063/5.0087894>

I. INTRODUCTION

Atomic clocks have become an indispensable technology for realizing a global navigation satellite system (GNSS) and timing synchronization of high-speed, high-capacity communications. They also enable precise measurement of physical quantities and have supported our understanding of physics.¹ For over 50 years, the unit of time, the second, has been defined by a cesium clock based on the microwave transition (≈ 9.2 GHz). Atomic fountain clocks, which employ laser-cooled atoms, have achieved a fractional uncertainty of 10^{-16} and give the best representation of the International System of Units (SI) second.²

Advances in laser technology, such as optical frequency combs,^{3,4} narrow-linewidth lasers,^{5,6} and frequency transfer technique using optical fibers,^{7,8} have allowed the dramatic progress of atomic clocks based on optical transitions. So far, single-ion clocks⁹ and optical lattice clocks^{10–12} have demonstrated a fractional uncertainty at 10^{-18} or below. Such optical clocks have outperformed the cesium clocks by two orders of magnitude, which facilitates the discussion for the redefinition of the second with optical clocks.

An optical clock with an uncertainty of 10^{-18} resolves the time dilation that corresponds to a 1 cm height difference on the ground. According to Einstein's theory of relativity, the gravitational redshift $\Delta\nu = \nu_2 - \nu_1$ between the two clocks is given by $\Delta\nu/\nu_1 = g\Delta h/c^2$, where c is the speed of light, g is the gravitational acceleration, and Δh is the height difference between the two clocks. Therefore, the height

difference between two sites can be measured by the frequency difference of the clocks set at the sites, which is referred to as the chronometric leveling.¹³ In Europe, for example, an optical fiber network over 2000 km is established to connect optical clocks developed in national metrology institutes.⁸ Such networked clocks may serve as the references for the geopotential or the height over the continent in the future.

So far, the state-of-the-art clocks with an uncertainty of 10^{-18} were demonstrated only in a well-controlled laboratory environment. In order to exploit atomic clocks outside the laboratories, transportable clocks are being developed. A group at the Physikalisch-Technische Bundesanstalt (PTB) developed an optical lattice clock with an uncertainty of 7.4×10^{-17} inside a car trailer.¹⁴ The clock was transported to the middle of a mountain in the Alpine massif and compared with the clocks at Istituto Nazionale di Ricerca Metrologica (INRiM) via an optical fiber link to demonstrate chronometric leveling.¹⁵ Such a transportable clock can also be used to compare clocks between national metrology institutes to confirm the consistency of clock frequency, which is one of the essential cornerstones toward the redefinition of the second by optical clocks.¹⁶ The European Space Agency (ESA) plans a space mission (I-SOC) to operate an optical lattice clock on the International Space Station (ISS) with the target instability and uncertainty of the low 10^{-17} level. A demonstrator model was developed with a fractional uncertainty of 2×10^{-17}

(Ref. 17). Optical clocks also provide a platform for testing fundamental physics. Some theoretical predictions speculate that dark matter could be probed using a global network of atomic clocks.¹⁸ It is also proposed that not only the transient signatures but also spatial variations of fundamental constants may serve as probes of dark matter,¹⁹ which can be studied with transportable optical clocks.

Other than static phenomena so far investigated by clocks, an excellent stability of optical lattice clocks accessing a fractional uncertainty of 10^{-18} in less than an hour may be used to observe dynamic behavior such as crustal movements and volcanic activities.²⁰ In particular, since the Japanese Island Arc, which is located at the plate boundary, is subject to earthquakes and volcanic activities, geodesy in Japan has focused on investigating minute temporal as well as spatial variations in Earth's crust.²¹

We first introduce the principle of optical lattice clocks and describe transportable clocks that show the performance competitive to laboratory-based clocks (Sec. II). We discuss the applications of fiber-linked optical lattice clocks in chronometric leveling (Sec. III), where the stability of optical lattice clocks allows the observation of not only the static but also the dynamic variation of geopotential.

II. DEVELOPMENT OF TRANSPORTABLE OPTICAL LATTICE CLOCKS

A. Principles of optical lattice clocks and choice of atomic element

A standing wave of lasers provides a periodic trapping potential with ultracold atoms. An array of atoms, thus, created is called an optical lattice. In the lattice, atoms are confined in a region smaller than the wavelength of a probe laser, which is referred to as the Lamb-Dicke regime. Such a confinement allows high-precision spectroscopy of atoms by suppressing the Doppler as well as the recoil shifts.²² However, the lattice trap created by the a.c. Stark shift of the atomic state may affect the transition frequencies of atoms.

The a.c. Stark shift in the i -state caused by the electric field $\mathbf{E}(\omega, \mathbf{r})$ with frequency ω and position $\mathbf{r}$ is given by

$$U_i \approx -\frac{1}{2} \alpha_i^{E1}(\omega, \boldsymbol{\varepsilon}) |\mathbf{E}(\omega, \mathbf{r})|^2,$$

where $\alpha_i^{E1}(\omega, \boldsymbol{\varepsilon})$ is the dipole polarizability that depends on frequency ω and polarization $\boldsymbol{\varepsilon}$ of the electric field. For the transition between

the $i = 1 \rightarrow 2$ states of atoms trapped in the lattice potential, the transition frequency ν is given by

$$h\nu \approx h\nu_0 - \frac{1}{2} [\alpha_2^{E1}(\omega, \boldsymbol{\varepsilon}) - \alpha_1^{E1}(\omega, \boldsymbol{\varepsilon})] |\mathbf{E}(\omega, \mathbf{r})|^2,$$

where ν_0 is the eigen frequency of an atom and h denotes the Planck constant. If $\alpha_1^{E1}(\omega_m, \boldsymbol{\varepsilon}_m) = \alpha_2^{E1}(\omega_m, \boldsymbol{\varepsilon}_m)$ holds, the a.c. Stark shift can be canceled on the clock transition²³ except the higher order terms of atom-field interactions. Here, ω_m is called the magic frequency, and the treatment for the higher order terms is described elsewhere.²⁴

While the frequency ω_m can be accurately measured, the light polarization $\boldsymbol{\varepsilon}_m$ is difficult to measure precisely. To suppress the polarization effects,²⁵ the states with the total angular momentum $J=0$ can be conveniently used as the clock states: this leads to the proposal of optical lattice clocks on the $^1S_0 - ^3P_0$ transition of alkaline-earth-metal-like atoms.²⁵ As the transition between $J=0$ states is strictly forbidden, isotopes with non-zero nuclear spins are used to couple the 3P_0 state with the $J=1$ state, such as the 1P_1 and 3P_1 states, to introduce transition moment on the clock transition. Based on this procedure, optical lattice clocks have been developed with strontium (Sr),^{10,11} ytterbium (Yb),¹² mercury (Hg),^{26,27} cadmium (Cd),²⁸ and magnesium (Mg)²⁹ atoms. Recently, an optical lattice clock using an inner-shell magnetic dipole transition of thulium (Tm) atoms with $J \neq 0$ was proposed.³⁰

Candidate atomic species are listed in Table I. The laser wavelengths required for Sr and Yb based clocks are in the visible to near-infrared region, where semiconductor diode lasers are readily available. On the other hand, those for Hg and Cd atoms are in the deep ultraviolet (DUV) region. From the viewpoint of designing transportable clocks, Sr and Yb atoms have an advantage in the availability of laser sources that allow for robust and stable laser systems to be fit into relatively small sizes.

A drawback for the Sr and Yb based clocks is their susceptibility to blackbody radiation (BBR) (Table I). The BBR shift is the a.c. Stark shift caused by blackbody radiation emitted from the surrounding environment. At a room temperature of 300 K, the BBR shift amounts to -5.3×10^{-15} (Ref. 31) and -2.7×10^{-15} (Ref. 32) for Sr and Yb atoms. This BBR shift is strongly dependent on the ambient temperature T as T^4 following the Stefan-Boltzmann law. To reduce

TABLE I. Clock parameters for candidate atomic species.

Atomic species	²⁵ Mg	⁸⁷ Sr	¹¹¹ Cd	¹⁹⁹ Hg	¹⁶⁹ Tm	¹⁷¹ Yb
Nuclear spin	5/2	9/2	1/2	1/2	1/2	1/2
Clock λ (nm)	458	698	332	266	1140	578
Magic λ (nm)	468	813	420	361	813,1064	759
First cooling λ (nm)	285	461	229	185	411	399
Second cooling λ (nm)	457	689	326	254	531	556
Temperature ^a (μ K)	1.8	0.2	1.6	31	9	4.4
BBR shift ^b (10^{-16})	-3.9 (Ref. 29)	-53 (Ref. 31)	-2.8 (Ref. 28)	-1.6 (Ref. 37)	0.023 (Ref. 30)	-27 (Ref. 32)
Hyperpolarizability ³³ $\left[\frac{\mu\text{Hz}}{(\text{kW}/\text{cm}^2)^2} \right]$	...	-200	-5	-3	...	-312

^aAtomic temperature after the second stage cooling.

^bBBR shift at 300 K.

this uncertainty, we developed a cryogenic clock equipped with a cold thermal shield for atoms.¹¹ The uncertainty of the BBR shift was reduced to 9×10^{-19} , which was limited by that of the temperature measurement at 95 K. The consistency of two cryogenic clocks was verified with 4.4×10^{-18} uncertainty.

Another source of uncertainty is the higher-order terms of atom-lattice interactions.^{24,33} While the a.c. Stark shift introduced by the dipole polarizability can be canceled by tuning the lattice laser to the magic frequency as described earlier, the higher-order Stark shift caused by the hyperpolarizability and multipolar polarizability effects remains with the uncertainty of 10^{-16} . This lattice light shift has been experimentally investigated with Sr,^{34,35} Yb,³⁶ and Hg²⁷ atoms (Table I). An operational magic condition was proposed to reduce the total Stark shift arising from electric dipole polarizability, hyperpolarizability, and multipolar polarizability by tuning the frequency and intensity of the lattice laser,²⁴ which has been experimentally investigated for Yb³⁶ and Sr³⁵ atoms.

B. Stability of optical lattice clocks

Optical atomic clocks employ a narrow linewidth laser, which is stabilized to a reference cavity, to excite an optical transition of atoms at the frequency ν_0 . The excited fraction of atoms is measured to servo-control the laser frequency to be resonant with the clock transition. The residual fluctuation of the, thus, stabilized laser frequency can be expressed by the clock instability. If all other sources of noise are negligible, the instability of the atomic clock follows the quantum projection noise (QPN) limit described by $\sigma_y^{(\text{QPN})}(\tau) \approx \frac{\Delta\nu}{\nu_0} \frac{1}{\sqrt{n\tau/T_c}}$,

which improves by probing the narrow spectrum $\Delta\nu$ for a large number of atoms n , and by extending the averaging time τ . Here, n denotes the number of atoms probed in a cycle time T_c of the measurement.

Compared with single ion clocks, optical lattice clocks probing a large number of atoms may reduce the averaging time by a factor of n . Figure 1 shows the QPN-limited instabilities for comparing two clocks with $n = 1$ (blue line) and 1000 (red line), respectively. Consequently, shorter averaging time required to reach a target uncertainty makes optical lattice clocks useful devices for observing time-varying events (Sec. III).

In reality, however, the instability of the clocks is limited by the instability of the clock laser. The periodic interrogation of the clock transition with a dead time down-samples the high frequency component of laser frequency noise and deteriorates the clock instability, which is called the Dick effect.³⁸ Let us assume the clock laser instability with a flicker floor of 1×10^{-16} at 0.1–10 s (green dashed line), the Dick effect limits the instability of clock comparison to $1.1 \times 10^{-16}/\sqrt{\tau/s}$ (brown line), which is equivalent to the QPN-limited instability for $n \approx 100$. Therefore, the potential stability of the optical lattice clocks is not fully exploited.

One straightforward way to reduce the clock instability is to develop a stable clock laser. As the instability of the clock laser is fundamentally limited by the thermal noise of cavity mirrors, half a meter long cavities³⁹ and a colder cavity have been developed to mitigate the thermal noise. A cryogenic silicon cavity^{6,40} was developed to achieve the laser instability of 2.8×10^{-17} at 1 s averaging time (green dashed line). Using such an ultrastable laser, the clock instability of $4.8 \times 10^{-17}/\sqrt{\tau/s}$ was demonstrated.⁴⁰

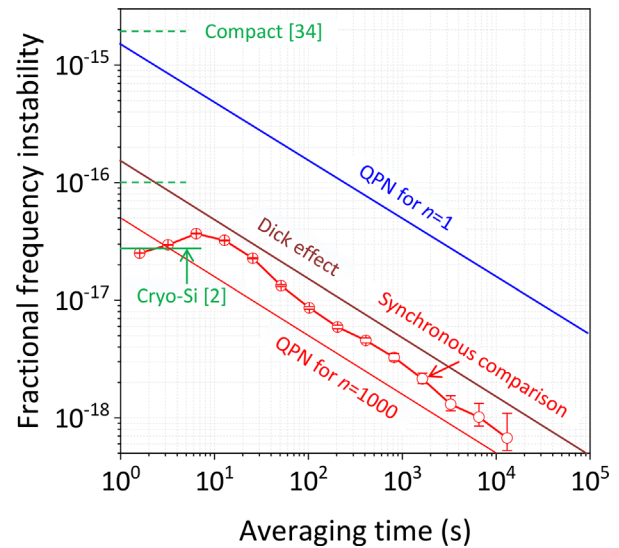


FIG. 1. Fractional frequency instabilities of clock comparison and cavity-stabilized lasers. The quantum projection noise limit is calculated for $\Delta\nu = 1.2$ Hz, $\nu_0 = 429$ THz, and $T_c = 1.6$ s. The blue and red lines show $\sqrt{2}\sigma_y^{(\text{QPN})}$ for $n = 1$ and $n = 1000$, respectively. The instability introduced by the Dick effect (brown line) is calculated for the laser instability at a flicker floor of 1×10^{-16} at 0.1–10 s. Frequency comparison of two synchronously operated clocks (red circles) partially rejects the frequency noise of the interrogation laser and reduces the instability close to the QPN limit. The instability of the laser stabilized to a cryogenic silicon cavity is shown by green line.

In general, such a highly stable laser system is not fully compatible with transportability, as it requires a long cavity that is sensitive to vibrational noise or a rather complicated cryogenic cooling system. The instability of 3×10^{-16} at 1 s was demonstrated with a transportable probe laser by designing a cavity insensitive to vibrational perturbation.⁴¹

On the other hand, several schemes were proposed to improve the stability without relying on the reference cavity. Operating two clocks by sharing a single clock laser reduces the instability of the frequency comparison beyond the Dick-effect limit⁴² as shown by red circles in Fig. 1. Such synchronous comparison speeds up the evaluation of the frequency difference between the clocks and can be applied to chronometric leveling.^{43,44} In order to improve the clock stability, zero-deadtime interrogation,⁴⁵ nondestructive measurement with atom recycling,^{46,47} super-radiant lasers,⁴⁸ phase-locking scheme,⁴⁹ and continuous operation with longitudinal Ramsey spectroscopy⁵⁰ were proposed. The zero-deadtime operation allows the clock stability to improve as $\sigma_y \propto 1/\tau$, which drastically shortens the averaging time until it reaches the QPN-limited instability. Assuming a laser instability of 2×10^{-15} at 0.1–1 s, which is realized by using a compact transportable cavity,⁵¹ the clock instability can reach low 10^{-19} with an averaging time of a few 1000 s (Ref. 50). Such a scheme will open an opportunity to approach beyond 10^{-18} in tens of minutes where the conventional geodetic measurement cannot access, as discussed in Sec. III.

C. Transportable optical lattice clocks with Sr atoms

We have developed a pair of transportable optical lattice clocks with ^{87}Sr atoms⁵² [Fig. 2(a)]. Each clock consists of a physics package

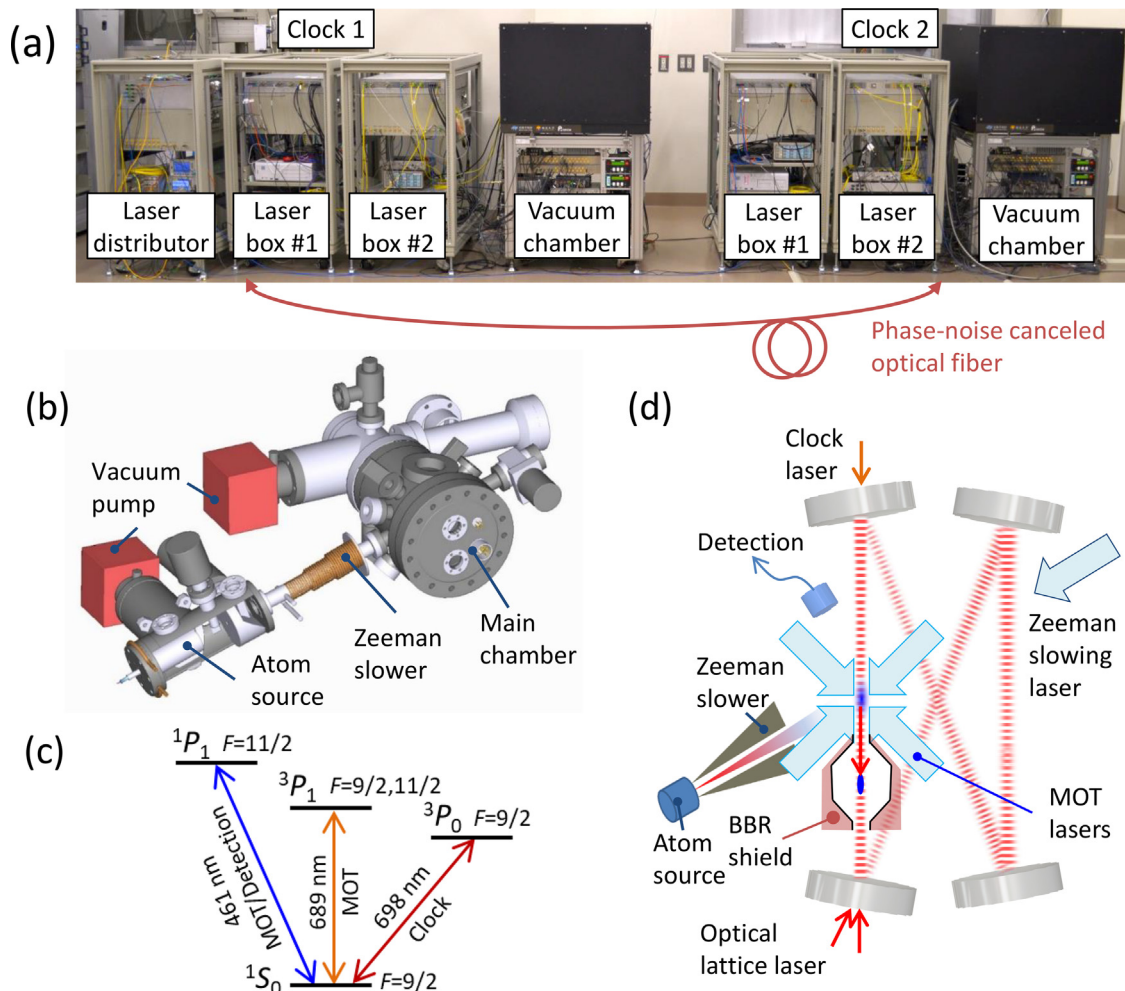


FIG. 2. (a) A pair of transportable clocks connected by a noise-canceled optical fiber. Each clock consists of a physics package and two laser boxes. Laser distributor box provides reference lasers for both clocks. (b) A design of the ultrahigh-vacuum chamber. The whole setup is set inside a magnetically shielded box with a size of $75(\text{W}) \times 55(\text{H}) \times 70(\text{D}) \text{ cm}^3$. (c) Relevant transitions of ^{87}Sr atoms. (d) A schematic of the physics package inside a vacuum chamber. Magneto-optically trapped ultracold ^{87}Sr atoms are loaded into a one-dimensional optical lattice that is formed inside a bow-tie cavity with four mirrors. Atoms are transferred into a BBR shield by a moving lattice and interrogated on the clock transition.

[Fig. 2(b)] and two laser boxes that house nine diode lasers used for cooling, lattice, and clock spectroscopy. The laser boxes are set in 19-in. racks, where the upper part of the box contains the lasers and optics and the lower part contains control electronics.

The ^{87}Sr atoms effused from an atomic oven are decelerated in the Zeeman slower and are magneto-optically trapped on the 1S_0 ($F = 9/2$) $\rightarrow$ 1P_1 ($F = 11/2$) transition [Fig. 2(c)]. The atoms are further cooled down to a few μK by the second-stage magneto-optical trap (MOT) on the narrow-line 1S_0 ($F = 9/2$) $\rightarrow$ 3P_1 ($F = 9/2, 11/2$) transition and are loaded into a one-dimensional optical lattice formed inside a bow-tie cavity [Fig. 2(d)], where the counter-propagating lattice lasers are tuned to the magic wavelength at 813 nm. Using a moving lattice, the atoms are transferred into a thermal shield to interrogate the clock transition 1S_0 ($F = 9/2$) $\rightarrow$ 3P_0 ($F = 9/2$). The shield is temperature-stabilized at 245 K by a four-stage Peltier cooler to reduce the BBR shift uncertainty to 2.6×10^{-18} , which is limited by the temperature measurement

uncertainty. After the interrogation, the excitation fraction of atoms is measured by the electron shelving technique⁵³ on the $^1S_0 \rightarrow ^1P_1$ transition, which is used to steer the clock laser frequency to be resonant with the clock transition. By repeating the clock cycles that consist of cooling and trapping of atoms in the optical lattice and of interrogating atoms, the statistical uncertainty of frequency comparison is reduced. For a synchronous comparison, we typically achieve an instability of 1×10^{-18} in 2 h as shown by the red circles in Fig. 1.

A pair of clocks connected by a phase-noise canceled optical fiber [Fig. 2(a)] can be used to measure the gravitational potential difference. We have measured gravitational redshift with a height difference of $\Delta h \sim 450 \text{ m}$ in a broadcasting tower, Tokyo Skytree,⁴⁴ by setting one of the transportable clocks in the ground floor and the other in the observatory floor at 450 m. The frequency difference between the clocks was measured with an uncertainty of 4.3×10^{-18} . Combined with geodetically measured gravitational potential differences between

the clocks, we have tested the gravitational redshift with the uncertainty of 9.1×10^{-5} , which was comparable to that of the recent space experiments.^{54,55} The measurement uncertainty of the gravitational redshift yields the height uncertainty of 4 cm, which indicates that the optical lattice clock is technically ready for chronometric leveling with centimeter-level uncertainty.

III. GEODETIC APPLICATIONS OF FIBER-LINKED OPTICAL LATTICE CLOCKS

A. Chronometric leveling

In geodesy, a height above sea level (orthometric height) is defined as $h(\mathbf{r}) = W(\mathbf{r})/g$, where $W(\mathbf{r})$ denotes the gravitational potential at a

position $\mathbf{r}$, relative to that at a mean sea level, and $g (\simeq 9.8 \text{ ms}^{-2})$ is an average surface gravity⁵⁶ [Fig. 3(a)]. Conventionally, W has been determined by integrating the potential increments that are obtained by multiplying a height difference observed with geodetic leveling with a mean gravity value along the observation route ($W(\mathbf{r}) = \sum_i g_i \Delta h_i$). Geodetic leveling takes considerable measurement time (typically, 2 km/day) and costs. Furthermore, the measurement uncertainty deteriorates for a longer-distance survey. In Japan, as an example, the allowable error for the first-order leveling survey is defined as $2.5 \sqrt{S}/\text{km mm}$ with total length of the route S , and observational results exceeding this limit are discarded.

In contrast, fiber-linked two distant optical clocks can measure the gravitational potential difference between the clock sites much

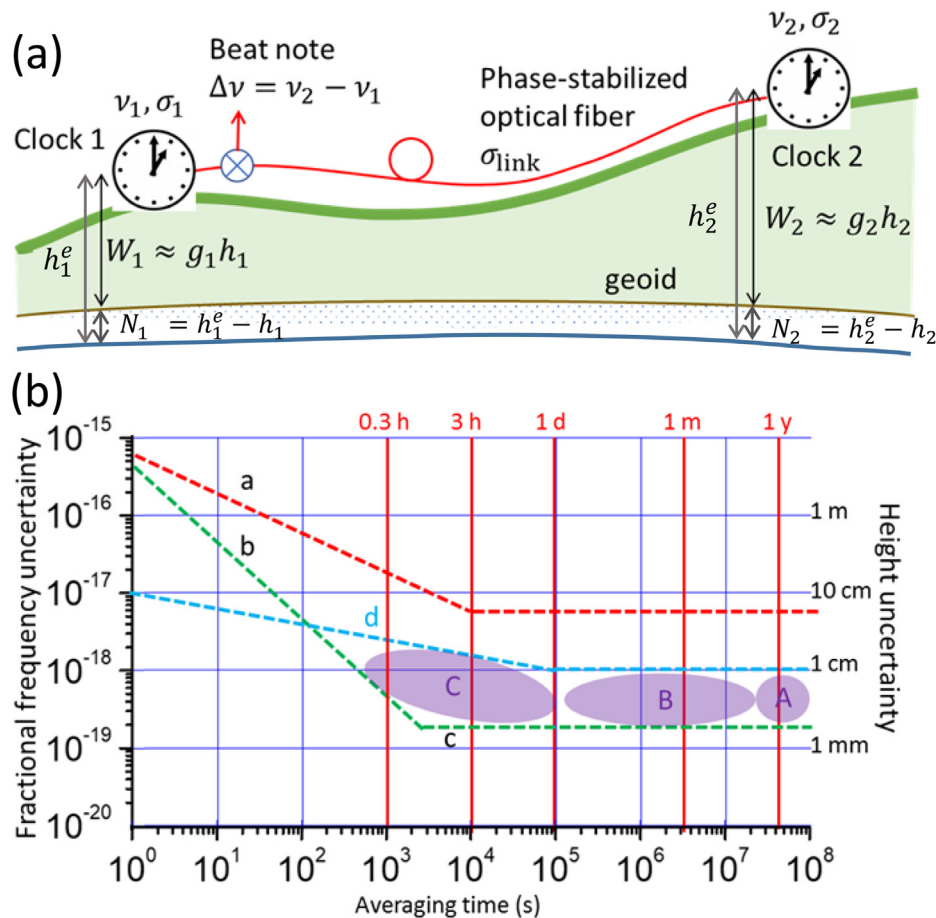


FIG. 3. (a) A schematic for chronometric leveling. On sites 1 and 2, the gravitational potential is given by W_1 and W_2 , and the clock frequency with gravitational redshift are given by ν_1 and ν_2 , respectively. The clock signal is transferred by a phase-stabilized optical fiber (red line). The beat note of the clocks $\Delta\nu = \nu_2 - \nu_1$ measures the gravitational potential difference $\Delta W = W_2 - W_1$ between two sites as $\Delta\nu/\nu_1 = \Delta W/c^2$. Assuming the gravitational acceleration $g_1 = g_2 = 9.8 \text{ m/s}^2$, the height difference $\Delta h = h_2 - h_1$ is given by $\Delta h = \frac{c^2 \Delta\nu}{g \nu_1} \approx 9.2 \times 10^{17} \frac{\Delta\nu}{\nu_1} (\text{cm})$. The gravitational potential W_i is measured from the reference equipotential surface marked by “geoid.” The curve below the geoid represents the reference ellipsoid, which serves as the reference for positioning with space geodetic techniques such as GNSS. The height h_i , which is measured with spirit leveling, is called the orthometric height. It is related to the ellipsoidal height h_i^e , which is measured with GNSS, as $h_i^e = h_i - N_i$, where N_i represents the geoid height. (b) Fractional frequency uncertainties for fiber-linked optical lattice clocks as a function of an averaging time τ . Right axis gives a corresponding height uncertainty. a—Frequency comparison of clocks reported in Ref. 41. b—Future optical clocks with zero-deadtime operation will improve the stability as τ^{-1} . c—Operational magic condition may improve the clock uncertainty to low 10^{-19} . d—A typical uncertainty for GNSS height measurement. (A)–(C) represent events which are expected to be (more precisely) detected by using the fiber-linked optical lattice clocks. (A) seasonal crustal movements including slow slip events (SSEs) and the secular deformation, (B) SSEs, and (C) solid Earth and ocean tides, SSEs, which are not yet discovered, possible slow slip prior to great earthquakes and volcanic eruptions with shorter duration than one day.

more quickly ($\leq$ several hours), based on the gravitational redshift, with no significant deterioration of the measurement uncertainty for longer distances [Fig. 3(a)].²¹ This means that optical clocks allow a more effective and precise determination of the height difference (chronometric leveling).⁵⁷

B. Chronometric leveling of static potential difference

Height data obtained with the conventional method based on geodetic leveling are combined with other geodetic data (position, gravity, etc.) to construct a height reference system, prescribing the origin of height in a country/region,⁵⁷ which is socially important. However, as mentioned earlier, geodetic leveling takes time and costs, and the mean sea level in a region also changes in longer time scales. Consequently, the zero-height surface officially adopted in a country/region, which was determined from observation data obtained during a certain time span, does not necessarily agree with those adopted in other countries/regions. To establish a common, global height reference system is one of the ultimate goals of geodesy, for which an effective measurement method to determine the gravitational potential is required. Fiber-linked optical lattice clocks serve for this purpose. Some earlier results in the application of optical clocks to a height reference system are found in Ref. 57.

C. Dynamic potential difference

For constructing a height reference system, only a static component of the gravitational potential is used because a static reference is more convenient to most users such as surveyors. However, the actual gravitational potential is disturbed by tides and variations in atmospheric pressure, continental water and non-tidal ocean bottom pressure, ice mass changes, great earthquakes, and so on.⁵⁸ Longer-wavelength (>200 km) variations in the gravitational potential due to these phenomena have already been monitored with satellite missions globally with a temporal resolution of one month.⁵⁹ A network of fiber-linked optical clocks on Earth's surface enables such dynamic potential changes to be observed with a better spatial as well as temporal resolution, which will provide new data for geophysical studies.

An application making the most of superb stability of optical lattice clocks is detection of geopotential changes caused by crustal deformations associated with earthquakes and volcanic activities. For such tectonic variations, temporal changes in the potential difference or the gravitational redshift observed by fiber-linked clocks are dominated by the height changes. That is, the potential change due to mass redistributions is negligibly small compared to the effect of height change at the clock sites.^{20,21}

In geodesy, temporal variations in the ellipsoidal height⁵⁶ [Fig. 3(a)], which is different from height observed by geodetic leveling, have been monitored by continuous GNSS observation networks, deployed around the world. Crustal movement data from GNSS have been widely used for studying earthquake and volcanic phenomena. Positioning uncertainty of GNSS depends on averaging time, which is typically 1 cm (a few mm) for the vertical (horizontal) coordinate for an averaging time of 24 h [Fig. 3(b)]. The larger uncertainty for vertical positioning is mainly due to incomplete modeling of atmospheric noise.⁶⁰

Figure 3(b) shows an uncertainty expected by chronometric leveling using future optical lattice clocks and GNSS height measurement. It might seem strange to compare the ellipsoidal height measured by

GNSS with the gravitational potential measured by chronometric leveling as they are different quantities. As mentioned before, however, potential changes caused by earthquakes and volcanic activities are dominated by height (or orthometric height) changes. The ellipsoidal height is related to the orthometric height as $h^e = h + N$, where N denotes the geoid height. The time derivative of this equation gives $dh^e/dt = dh/dt + dN/dt$. The dominance of height changes, i.e., $dN/dt \ll dh/dt$, validates the above direct comparison between the ellipsoidal height and the orthometric height in the sense of temporal change. Figure 3(b) indicates that chronometric leveling will be able to detect dynamic height changes faster than GNSS for some time scales. Generally, earthquakes and volcanic activities cause not only horizontal but also vertical deformation. An accurate measurement of the three-dimensional deformation pattern is important to identify the cause of the observed deformation. Combination of height monitoring with optical lattice clocks and horizontal deformation monitoring with GNSS realizes such a measurement (Fig. 4). The combined observation will improve identification of the precursory slip to a great earthquake and a rapid magma intrusion [Fig. 3(b-C)], short-term fault slip events on a plate boundary [Figs. 3(b-B) and 3(b-C)], and seasonal deformations [Fig. 3(b-A)], for example, see Ref. 21.

As a first step to realize such an application of optical lattice clocks to the detection of crustal movements, a 100-km-long fiber-link was established in 2020.⁶¹ The fiber link around Tokyo will be further extended to northeastern Japan by a 700-km-long fiber, where we plan to investigate the ongoing post-seismic deformation triggered by the 2011 M9 earthquake.⁶² As it is not predictable in general when earthquake will occur, extending the operation time of clocks is important for applications in seismology.

IV. SUMMARY

The high accuracy and stability of optical lattice clocks make them ideal devices for measuring gravitational potential; their accuracy can be used as a reference for the altitude, and their stability allows observing the dynamic behavior of the gravitational potential. We have developed a transportable Sr optical lattice clock suitable to these applications; our clocks were transported and tested at Tokyo SkyTree

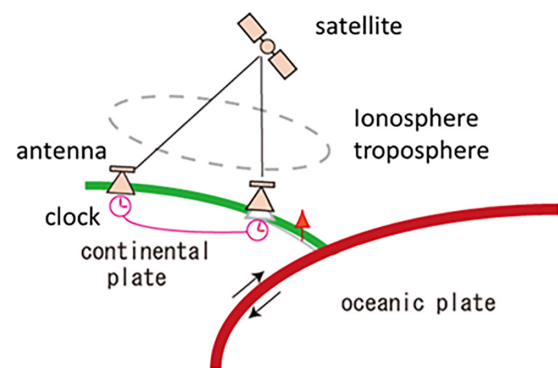


FIG. 4. Schematic representation of the crustal deformation monitoring using fiber-linked optical lattice clocks. Vertical motion is masked by atmospheric noise in conventional GNSS observation. By combining chronometric leveling with GNSS observation, we can detect vertical deformation more quickly and precisely. Improved crustal deformation data with significantly reduced atmospheric noise allow the fault slip on the plate subduction interface to be inferred more clearly.

to operate with an uncertainty of 4×10^{-18} . Such clocks may be used to investigate geophysical applications by setting the clocks over a long baseline connected by an optical fiber and by comparing the clocks. By applying new schemes such as zero-deadtime operation, chronometric leveling at or below a centimeter of uncertainty can be done in a time-scale of less than an hour. Such precise geodetic measurement in real time is beyond the capability of conventional spirit leveling or GNSS height determination and will open up new applications of chronometric leveling. Further miniaturization and improved robustness and durability of the clock will extend the applications to space experiment^{17,63} to investigate new physics¹ and laws of the universe.

ACKNOWLEDGMENTS

This work was supported by the JST-Mirai Program (Grant No. JPMJMI18A1) and the JSPS Grant-in-Aid for Specially Promoted Research (Grant No. JP16H06284).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

REFERENCES

- M. S. Safronova, D. Budker, D. DeMille, D. F. J. Kimball, A. Derevianko, and C. W. Clark, "Search for new physics with atoms and molecules," *Rev. Mod. Phys.* **90**, 025008 (2018).
- C. R. Physique **20**, 153–168 (2019).
- T. Udem, R. Holzwarth, and T. W. Hänsch, "Optical frequency metrology," *Nature* **416**, 233 (2002).
- S. T. Cundiff and J. Ye, "Colloquium: Femtosecond optical frequency combs," *Rev. Mod. Phys.* **75**, 325–342 (2003).
- B. C. Young, F. C. Cruz, W. M. Itano, and J. C. Bergquist, "Visible lasers with subhertz linewidths," *Phys. Rev. Lett.* **82**, 3799–3802 (1999).
- D. G. Matei, T. Legero, S. Hafner, C. Grebing, R. Weyrich, W. Zhang, L. Sonderhouse, J. M. Robinson, J. Ye, F. Riehle, and U. Sterr, "1.5 μm lasers with sub-10 mHz linewidth," *Phys. Rev. Lett.* **118**, 263202 (2017).
- L.-S. Ma, P. Jungner, J. Ye, and J. L. Hall, "Delivering the same optical frequency at two places: Accurate cancellation of phase noise introduced by an optical fiber or other time-varying path," *Opt. Lett.* **19**, 1777–1779 (1994).
- F. Riehle, "Optical clock networks," *Nat. Photonics* **11**, 25–31 (2017).
- S. M. Brewer, J. S. Chen, A. M. Hankin, E. R. Clements, C. W. Chou, D. J. Wineland, D. B. Hume, and D. R. Leibrandt, "²⁷Al⁺ quantum-logic clock with a systematic uncertainty below 10^{-18} ," *Phys. Rev. Lett.* **123**, 033201 (2019).
- T. L. Nicholson, S. L. Campbell, R. B. Hutson, G. E. Marti, B. J. Bloom, R. L. McNally, W. Zhang, M. D. Barrett, M. S. Safronova, G. F. Strouse, W. L. Tew, and J. Ye, "Systematic evaluation of an atomic clock at 2×10^{-18} total uncertainty," *Nat. Commun.* **6**, 6896 (2015).
- I. Ushijima, M. Takamoto, M. Das, T. Ohkubo, and H. Katori, "Cryogenic optical lattice clocks," *Nat. Photonics* **9**, 185–189 (2015).
- W. F. McGrew, X. Zhang, R. J. Fasano, S. A. Schaffer, K. Beloy, D. Nicolodi, R. C. Brown, N. Hinkley, G. Milani, M. Schioppo, T. H. Yoon, and A. D. Ludlow, "Atomic clock performance enabling geodesy below the centimetre level," *Nature* **564**, 87–90 (2018).
- A. Bjerhammar, "On a relativistic geodesy," *Bull. Géod.* **59**, 207–220 (1985).
- S. B. Koller, J. Grotti, S. Vogt, A. Al-Masoudi, S. Dörscher, S. Häfner, U. Sterr, and C. Lisdat, "Transportable optical lattice clock with 7×10^{-17} uncertainty," *Phys. Rev. Lett.* **118**, 073601 (2017).
- J. Grotti, S. Koller, S. Vogt, S. Häfner, U. Sterr, C. Lisdat, H. Denker, C. Voigt, L. Timmen, A. Rolland, F. N. Baynes, H. S. Margolis, M. Zampaolo, P. Thoumany, M. Pizzocaro, B. Rauf, F. Bregolin, A. Tampellini, P. Barbieri, M. Zucco, G. A. Costanzo, C. Clivati, F. Levi, and D. Calonico, "Geodesy and metrology with a transportable optical clock," *Nat. Phys.* **14**, 437–441 (2018).
- F. Riehle, P. Gill, F. Arias, and L. Robertsson, "The CIPM list of recommended frequency standard values: Guidelines and procedures," *Metrologia* **55**, 188–200 (2018).
- S. Origlia, M. S. Pramod, S. Schiller, Y. Singh, K. Bongs, R. Schwarz, A. Al-Masoudi, S. Dörscher, S. Herbers, S. Häfner, U. Sterr, and C. Lisdat, "Towards an optical clock for space: Compact, high-performance optical lattice clock based on bosonic atoms," *Phys. Rev. A* **98**, 053443 (2018).
- A. Derevianko and M. Pospelov, "Hunting for topological dark matter with atomic clocks," *Nat. Phys.* **10**, 933–936 (2014).
- Y. V. Stadnik, "New bounds on macroscopic scalar-field topological defects from nontransient signatures due to environmental dependence and spatial variations of the fundamental constants," *Phys. Rev. D* **102**, 115016 (2020).
- M. Bondarescu, P. Jetzer, N. Houlié, G. Hetényi, A. Lundgren, A. Schäfer, and R. Bondarescu, "Ground-based optical atomic clocks as a tool to monitor vertical surface motion," *Geophys. J. Int.* **202**, 1770–1774 (2015).
- Y. Tanaka and H. Katori, "Exploring potential applications of optical lattice clocks in a plate subduction zone," *J. Geod.* **95**, 93 (2021).
- T. Ido and H. Katori, "Recoil-free spectroscopy of neutral Sr atoms in the Lamb-Dicke regime," *Phys. Rev. Lett.* **91**, 053001 (2003).
- H. Katori, M. Takamoto, V. G. Palchikov, and V. D. Ovsiannikov, "Ultrastable optical clock with neutral atoms in an engineered light shift trap," *Phys. Rev. Lett.* **91**, 173005 (2003).
- H. Katori, V. D. Ovsiannikov, S. I. Marmo, and V. G. Palchikov, "Strategies for reducing the light shift in atomic clocks," *Phys. Rev. A* **91**, 052503 (2015).
- H. Katori, *Proceedings of the 6th Symposium on Frequency Standards and Metrology*, edited by P. Gill (World Scientific, Singapore, 2002), p. 323.
- J. J. McFerran, L. Yi, S. Mejri, W. Zhang, S. Di Manno, M. Abgrall, J. Guéna, Y. Le Coq, and S. Bize, "Statistical uncertainty of 2.5×10^{-16} for the ¹⁹⁹Hg ¹S₀ – ³P₀ clock transition against a primary frequency standard," *Phys. Rev. A* **89**, 043432 (2014).
- K. Yamanaka, N. Ohmae, I. Ushijima, M. Takamoto, and H. Katori, "Frequency ratio of ¹⁹⁹Hg and ⁸⁷Sr optical lattice clocks beyond the SI limit," *Phys. Rev. Lett.* **114**, 230801 (2015).
- A. Yamaguchi, M. S. Safronova, K. Gibble, and H. Katori, "Narrow-line cooling and determination of the magic wavelength of Cd," *Phys. Rev. Lett.* **123**, 113201 (2019).
- A. P. Kulosa, D. Fim, K. H. Zipfel, S. Ruhmann, S. Sauer, N. Jha, K. Gibble, W. Ertmer, E. M. Rasel, M. S. Safronova, U. I. Safronova, and S. G. Porsev, "Towards a Mg lattice clock: Observation of the ¹S₀ – ³P₀ transition and determination of the magic wavelength," *Phys. Rev. Lett.* **115**, 240801 (2015).
- A. Golovizin, E. Fedorova, D. Tregubov, D. Sukachev, K. Khabarova, V. Sorokin, and N. Kolachevsky, "Inner-shell clock transition in atomic thulium with a small blackbody radiation shift," *Nat. Commun.* **10**, 1724 (2019).
- C. Lisdat, S. Dörscher, I. Nosske, and U. Sterr, "Blackbody radiation shift in strontium lattice clocks revisited," *Phys. Rev. Res.* **3**, L042036 (2021).
- K. Beloy, J. A. Sherman, N. D. Lemke, N. Hinkley, C. W. Oates, and A. D. Ludlow, "Determination of the ^{5d}6s ³D₁ state lifetime and blackbody-radiation clock shift in Yb," *Phys. Rev. A* **86**, 051404 (2012).
- V. D. Ovsiannikov, S. I. Marmo, V. G. Palchikov, and H. Katori, "Higher-order effects on the precision of clocks of neutral atoms in optical lattices," *Phys. Rev. A* **93**, 043420 (2016).
- P. G. Westergaard, J. Lodewyck, L. Lorini, A. Lecallier, E. A. Burt, M. Zawada, J. Millo, and P. Lemonde, "Lattice-Induced Frequency Shifts in Sr Optical Lattice Clocks at the 10^{-17} Level," *Phys. Rev. Lett.* **106**, 210801 (2011).
- I. Ushijima, M. Takamoto, and H. Katori, "Operational magic intensity for Sr optical lattice clocks," *Phys. Rev. Lett.* **121**, 263202 (2018).
- R. C. Brown, N. B. Phillips, K. Beloy, W. F. McGrew, M. Schioppo, R. J. Fasano, G. Milani, X. Zhang, N. Hinkley, H. Leopardi, T. H. Yoon, D. Nicolodi, T. M. Fortier, and A. D. Ludlow, "Hyperpolarizability and operational magic wavelength in an optical lattice clock," *Phys. Rev. Lett.* **119**, 253001 (2017).

- ³⁷H. Hachisu, K. Miyagishi, S. G. Porsev, A. Derevianko, V. D. Ovsiannikov, V. G. Pal'chikov, M. Takamoto, and H. Katori, "Trapping of neutral mercury atoms and prospects for optical lattice clocks," *Phys. Rev. Lett.* **100**, 053001 (2008).
- ³⁸G. Santarelli, C. Audoin, A. Makdissi, P. Laurent, G. J. Dick, and A. Clairon, "Frequency stability degradation of an oscillator slaved to a periodically interrogated atomic resonator," *IEEE Trans. Ultrason. Ferroelectr. Frequency Control* **45**, 887–894 (1998).
- ³⁹S. Hafner, S. Falke, C. Grebing, S. Vogt, T. Legero, M. Merimaa, C. Lisdat, and U. Sterr, " 8×10^{-17} fractional laser frequency instability with a long room-temperature cavity," *Opt. Lett.* **40**, 2112–2115 (2015).
- ⁴⁰E. Oelker, R. B. Hutson, C. J. Kennedy, L. Sonderhouse, T. Bothwell, A. Goban, D. Kedar, C. Sanner, J. M. Robinson, G. E. Marti, D. G. Matei, T. Legero, M. Giunta, R. Holzwarth, F. Riehle, U. Sterr, and J. Ye, "Demonstration of 4.8×10^{-17} stability at 1s for two independent optical clocks," *Nat. Photonics* **13**, 714–719 (2019).
- ⁴¹S. Hafner, S. Herbers, S. Vogt, C. Lisdat, and U. Sterr, "Transportable interrogation laser system with an instability of mod $\sigma_y = 3 \times 10^{-16}$," *Opt. Express* **28**, 16407–16416 (2020).
- ⁴²M. Takamoto, T. Takano, and H. Katori, "Frequency comparison of optical lattice clocks beyond the Dick limit," *Nat. Photonics* **5**, 288–292 (2011).
- ⁴³T. Takano, M. Takamoto, I. Ushijima, N. Ohmae, T. Akatsuka, A. Yamaguchi, Y. Kuroishi, H. Munekane, B. Miyahara, and H. Katori, "Geopotential measurements with synchronously linked optical lattice clocks," *Nat. Photonics* **10**, 662–666 (2016).
- ⁴⁴M. Takamoto, I. Ushijima, N. Ohmae, T. Yahagi, K. Kokado, H. Shinkai, and H. Katori, "Test of general relativity by a pair of transportable optical lattice clocks," *Nat. Photonics* **14**, 411–415 (2020).
- ⁴⁵M. Schioppo, R. C. Brown, W. F. McGrew, N. Hinkley, R. J. Fasano, K. Beloy, T. H. Yoon, G. Milani, D. Nicolodi, J. A. Sherman, N. B. Phillips, C. W. Oates, and A. D. Ludlow, "Ultrastable optical clock with two cold-atom ensembles," *Nat. Photonics* **11**, 48–52 (2017).
- ⁴⁶G. Vallet, E. Bookjans, U. Eismann, S. Bilicki, R. L. Targat, and J. Lodewyck, "A noise-immune cavity-assisted non-destructive detection for an optical lattice clock in the quantum regime," *New J. Phys.* **19**, 083002 (2017).
- ⁴⁷R. Hobson, W. Bowden, A. Vianello, I. R. Hill, and P. Gill, "Cavity-enhanced non-destructive detection of atoms for an optical lattice clock," *Opt. Express* **27**, 37099–37110 (2019).
- ⁴⁸M. A. Norcia, M. N. Winchester, J. R. K. Cline, and J. K. Thompson, "Superradiance on the millihertz linewidth strontium clock transition," *Sci. Adv.* **2**, e1601231 (2016).
- ⁴⁹R. Kohlhaas, A. Bertoldi, E. Cantin, A. Aspect, A. Landragin, and P. Bouyer, "Phase locking a clock oscillator to a coherent atomic ensemble," *Phys. Rev. X* **5**, 021011 (2015).
- ⁵⁰H. Katori, "Longitudinal Ramsey spectroscopy of atoms for continuous operation of optical clocks," *Appl. Phys. Express* **14**, 072006 (2021).
- ⁵¹J. Davila-Rodriguez, F. N. Baynes, A. D. Ludlow, T. M. Fortier, H. Leopardi, S. A. Diddams, and F. Quinlan, "Compact, thermal-noise-limited reference cavity for ultra-low-noise microwave generation," *Opt. Lett.* **42**, 1277–1280 (2017).
- ⁵²N. Ohmae, M. Takamoto, Y. Takahashi, M. Kokubun, K. Araki, A. Hinton, I. Ushijima, T. Muramatsu, T. Furumiyu, Y. Sakai, N. Moriya, N. Kamiya, K. Fujii, R. Muramatsu, T. Shiimado, and H. Katori, "Transportable strontium optical lattice clocks operated outside laboratory at the level of 10^{-18} uncertainty," *Adv. Quantum Technol.* **4**, 2100015 (2021).
- ⁵³H. G. Dehmelt, "Monoion oscillator as potential ultimate laser frequency standard," *IEEE Trans. Instrum. Meas.* **IM-31**, 83–87 (1982).
- ⁵⁴P. Delva, N. Puchades, E. Schonemann, F. Dilssner, C. Courde, S. Bertone, F. Gonzalez, A. Hees, C. Le Poncin-Lafitte, F. Meynadier, R. Prieto-Cerdeira, B. Sohet, J. Ventura-Traveset, and P. Wolf, "Gravitational redshift test using eccentric Galileo satellites," *Phys. Rev. Lett.* **121**, 231101 (2018).
- ⁵⁵S. Herrmann, F. Finke, M. Lulf, O. Kichakova, D. Puetzfeld, D. Knickmann, M. List, B. Rievers, G. Giorgi, C. Gunther, H. Dittus, R. Prieto-Cerdeira, F. Dilssner, F. Gonzalez, E. Schonemann, J. Ventura-Traveset, and C. Lammerzahl, "Test of the gravitational redshift with Galileo satellites in an eccentric orbit," *Phys. Rev. Lett.* **121**, 231102 (2018).
- ⁵⁶B. Hofmann-Wellenhof and H. E. Moritz, *Physical Geodesy*, 2nd ed. (Springer-Verlag, Wien, 2006).
- ⁵⁷P. Delva, H. Denker, and G. Lion, *Chronometric Geodesy: Methods and Applications, Relativistic Geodesy, Foundations and Applications* (Springer International Publishing, 2019).
- ⁵⁸C. Voigt, H. Denker, and L. Timmen, "Time-variable gravity potential components for optical clock comparisons and the definition of international time scales," *Metrologia* **53**, 1365–1383 (2016).
- ⁵⁹B. Wouters, J. A. Bonin, D. P. Chambers, R. E. M. Riva, I. Sasgen, and J. Wahr, "Grace, time-varying gravity, Earth system dynamics and climate change," *Rep. Prog. Phys.* **77**, 116801 (2014).
- ⁶⁰J. Boehm and H. Schuh, *Atmospheric Effects in Space Geodesy* (Springer Atmospheric Sciences, 2013).
- ⁶¹T. Akatsuka, T. Goh, H. Imai, K. Oguri, A. Ishizawa, I. Ushijima, N. Ohmae, M. Takamoto, H. Katori, T. Hashimoto, H. Gotoh, and T. Sogawa, "Optical frequency distribution using laser repeater stations with planar lightwave circuits," *Opt. Express* **28**, 9186–9197 (2020).
- ⁶²K. Wang, Y. Hu, and J. He, "Deformation cycles of subduction earthquakes in a viscoelastic Earth," *Nature* **484**, 327–332 (2012).
- ⁶³K. Bongs, Y. Singh, L. Smith, W. He, O. Kock, D. Świerad, J. Hughes, S. Schiller, S. Alighanbari, S. Origlia, S. Vogt, U. Sterr, C. Lisdat, L. Targat, J. Lodewyck, D. Holleville, B. Venon, S. Bize, G. P. Barwood, P. Gill, I. R. Hill, Y. B. Ovchinnikov, N. Poli, G. M. Tino, J. Stuhler, and W. Kaenders, "Development of a strontium optical lattice clock for the SOC mission on the ISS," *C. R. Phys.* **16**, 553–564 (2015).